

RealityDrop: A Framework Using Multimodal Superhuman Techniques to Manipulate Cross-Reality Virtual Content

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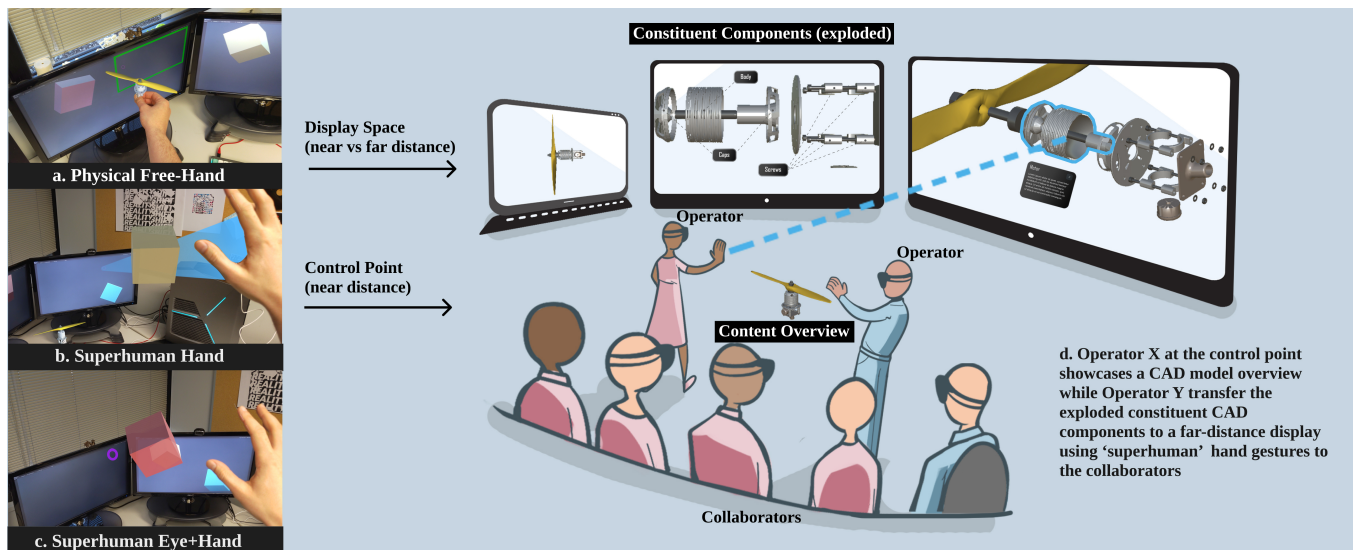


Figure 1: System overview and a use case

Abstract

We present RealityDrop, a novel framework using the “superhuman” multimodal interaction techniques to transfer, manipulate, and display cross-reality virtual content among domain experts. Employing Mixed Reality as the centre of control, RealityDrop affords a diverse background of multiusers (expert vs novice) to disseminate digital content (overview vs constituents) at varied control points to display space (near vs far) through multimodal superhuman interaction techniques (free hand vs hand+gaze). It builds an easy cross-system content transfer workflow that no longer requires the configuration and alignment of each display and environment.

1 customised content interpreter, 3 multimodal interaction techniques, and 2 cross-system transfer interfaces are incorporated for fluent content manipulation and presentation during collaboration.

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1 Introduction

Mixed Reality (MR) based display systems are introduced each year to improve collaboration in a shared workspace, enabling a vast opportunity for virtual content to be represented based on different levels of control, details and spatial features [2, 5]. These displays provide the affordance for a team of diverse experts to quickly navigate, present, and discuss information visualised in the MR task space freely [6]. 3D content faces challenges when displayed on conventional displays (desktop or mobile) due to a lack of intuitive

manipulation of its viewpoint, angles and substituent details between multiple users [4]. The associated interaction techniques are primarily based on mouse click-and-drag [7], while the transfer interface usually requires a close-distant display to minimise the duplications and effort [1, 3].

To overcome these challenges, we introduce **RealityDrop** (Figure 1), a novel multimodal MR framework to improve the workflow for multiusers to transfer and display cross-reality virtual content at different control points using natural free-hand interaction techniques. Our framework integrates four core concepts: (1) *A customized content interpreter* of two levels defines if a 3D overview (in MR control point) or its 'exploded' constituent components (in monitor display space) should be represented based on display capability; (2) *Three multimodal interaction techniques* include the basic free-hand drag-and-drop, 'superhuman' hand gestures, and "superhuman" gaze+hand to "drop" content using their gaze or palm "power" without holding a mouse or tablet to navigate; (3) In addition to conventional close-distance transfer, *cross-system transfer interfaces* enable users to stay at their current far-distance control point without physically walking up to a display or a collaborator to perform a cross-reality content transfer.

2 System Overview

In Figure 1: d, the workflow utilises different display specifications to accommodate the levels of content details that *operators* (i.e., the user who performs interactions at a specific control point) plan to present to the *collaborators*. For example, an operator who showcased a complete overview of an assembled computer-aided design (CAD) model subsequently dropped it from a distant *control point* into *display space* to collaborators using superhuman hand gestures, with each part of the model disseminated in exploded constituent components, from different angles and orientations.

2.1 Customised Content Interpreter

Virtual content is defined as 3D objects that can be transferred from a range of display scales (wall-size projectors and table-size touchscreens to smartphone and laptop screens) in two levels (*overview vs constituent components*). For example, bigger displays with higher resolution should reveal more 'exploded' constituent details, while the MR control point presents an overview of the content as a whole. The spectrum of content encompasses entities as simple as cubes to more intricate constructs like data visualisations or CAD models.

2.2 Multimodal Interaction Techniques

Physical Free-hand Drag-and-Drop is the baseline technique where the operator can physically grab and place the content between the control point and the display space using natural hand gestures (Figure 1: a).

Superhuman Hand uses in-air hand gestures, allowing users to manipulate content from afar. When interacting with the content, the operator raises their hand with palm forward, like a superhuman absorbing motion, towards a specific display. A ray will be highlighted from the centre of the operator's palm to 'pull/push' the content. The content will then 'fly' between the operator's hand in the MR control point and the display space. This technique requires no physical movement to a specific display and uses lines

connecting the user's palm to the chosen display (palm raycast) as directional cues to help with placement accuracy (Figure 1: b).

Superhuman Gaze+Hand selects the content using the user's eye gaze. The gaze cursor of the operator's current focus replaces the palm ray to indicate selected content, and the same superhuman gestures are then used to perform the corresponding 'pull/push' actions. This technique eases the operator's effort from raising hands to gazing at the content to select (Figure 1: c).

2.3 Cross-system Transfer Interface

Two spatial concepts are used to define the current positional state of the content. In the control point, the operator manipulates the arrangement of content in MR. In contrast, the display space denotes a specific display location (*near vs far*) where content is presented. As a result, two cross-system transfer interfaces, basic and distinct, are incorporated to add visual guidance and effects to the transfer process across realities from different locations, making the transfer more realistic from a 2D to an MR display (Figure 1: d).

Basic Transfer Interface translates the content's position and orientation between the release point and display locations to ensure consistent behaviours during transfer. The basic interface allows for simple adaptations by changing colour, scale and opacity between two locations. For example, by moving an object to a larger display from afar, the object could increase its scale but keep the colour to fit the dimensions of that display correctly. Meanwhile, the visual effects and spatial orientations of the virtual content's last interaction remain from one space to another.

Distinct Transfer Interface allows effects to be customised to a specific type of pre-defined content interpreter. For example, when operators placed content into different displays, a CAD model could transition from an assembled overview to multiple "exploded" views with the constituent component labels and details. This reinforces the audience's ability to understand the content of that display.

2.4 Software and Hardware

RealityDrop¹ was created in Unity 2021.3.15f1 and built for Microsoft HoloLens 2 and Windows. Any MR device using the Mixed Reality Toolkit (MRTK) or any display supporting Unity can be utilised by RealityDrop. The interaction component uses controller systems from the MRTK, and multiuser networking integration uses the Photon Unity Networking 2 (PUN2) plugin.

2.5 Use Case

Dissemination of meaningful information with a diverse range of experts using customised visualisations presents an exciting use case for RealityDrop. To illustrate, a designer, an engineer, and a PM can all perceive identical 3D content in distinct ways. The operator can use the custom content interpreter to manipulate MR virtual parts of a 3D CAD model and then present them to the audience in a manner that aligns with their expertise. Therefore, engineers view a mechanical schematic, designers focus on colours and text, while PM see an overarching summary (Figure 1: d).

¹<https://www.youtube.com/watch?v=CV098ngETjw>

3 Conclusion & Future Work

This poster proposes a multimodal MR framework to customise, transfer and display cross-reality virtual content. The operator interacts and manipulates virtual content in the MR control point, best amplifying (overview vs component) and transferring (near vs far) content to the corresponding display using ‘superhuman’ gaze or hand gestures, among diverse experts to collaborate. The cross-reality transfer workflow becomes intuitive and interactive across different display systems without complex on-site configuration. In the future, we plan to conduct a full user study to compare how interaction techniques (baseline, gestures, gaze+gestures) affect displayed information (customised and non-customised) among groups of domain experts.

References

- [1] H. Benko, E.W. Ishak, and S. Feiner. 2005. Cross-dimensional gestural interaction techniques for hybrid immersive environments. In *IEEE Proceedings. VR 2005. Virtual Reality, 2005*. 209–216. doi:10.1109/VR.2005.1492776
- [2] Andrew Cunningham, James Walsh, and Bruce Thomas. 2018. Immersive visualisation of geo-temporal narratives in law enforcement. In *2018 International Symposium on Big Data Visual and Immersive Analytics (BDVA)*. IEEE, 1–8.
- [3] Steven Feiner and Ari Shamash. 1991. Hybrid user interfaces: breeding virtually bigger interfaces for physically smaller computers. In *Proceedings of the 4th Annual ACM Symposium on User Interface Software and Technology* (Hilton Head, South Carolina, USA) (*UIST '91*). Association for Computing Machinery, New York, NY, USA, 9–17. doi:10.1145/120782.120783
- [4] Allison Jing, Chenyang Xiang, Seungwon Kim, Mark Billinghurst, and Aaron Quigley. 2019. SnapChart: An Augmented Reality Analytics Toolkit to Enhance Interactivity in a Collaborative Environment. In *Proceedings of the 17th International Conference on Virtual-Reality Continuum and Its Applications in Industry* (Brisbane, QLD, Australia) (*VRCAI '19*). Association for Computing Machinery, New York, NY, USA, Article 55, 2 pages. doi:10.1145/3359997.3365725
- [5] Benjamin Lee, Maxime Cordeil, Arnaud Prouzeau, Bernhard Jenny, and Tim Dwyer. 2022. A Design Space For Data Visualisation Transformations Between 2D And 3D In Mixed-Reality Environments. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (*CHI '22*). Association for Computing Machinery, New York, NY, USA, Article 25, 14 pages. doi:10.1145/3491102.3501859
- [6] Jeremy McDade, Adam Drogemuller, Allison Jing, Nick Ireland, James Walsh, Bruce Thomas, Wolfgang Mayer, and Andrew Cunningham. 2022. CADET: A Collaborative Agile Data Exploration Tool for Mixed Reality. In *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 899–900. doi:10.1109/ISMAR-Adjunct57072.2022.00195
- [7] Jun Rekimoto and Masanori Saitoh. 1999. Augmented Surfaces: A Spatially Continuous Work Space for Hybrid Computing Environments. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Pittsburgh, Pennsylvania, USA) (*CHI '99*). Association for Computing Machinery, New York, NY, USA, 378–385. doi:10.1145/302979.303113